Harshil Patel

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6+ years in data science & backend infrastructure; 3 years leading modeling, experimentation & production data platforms

PROFESSIONAL EXPERIENCE

Senior Data Scientist and R&D Lead | Miix Analytics | *ON, Canada* *Jul 2023 - Present*

- Led a ∼**15-person** data science team—hiring, mentoring, and raising the analytical bar—owning marketing measurement systems that steer **$50M–$80M** in annual media budgets for CPG, retail, and beverage clients across **Canada and the U.S.**
- **Owned Hive end-to-end** as product owner—roadmap, architecture, and delivery of a production MMM, forecasting, and attribution platform—turning bespoke consulting into a **scalable data product** serving multiple enterprise clients.
- Built **causal measurement at scale**: marketing mix models, **incrementality** and **attribution**, and **A/B and geo experiments** that quantified media ROI and reallocated spend for measurable lift.
- Productionized **Bayesian MMM with Google Meridian** (adstock/saturation, channel response, **budget/ROAS optimization**) on **TensorFlow/TFP**, delivering partner-ready models on client data.
- Architected Hive's production stack—**FastAPI**, distributed **Celery** workers, **Redis**, object storage, and **CI/CD on GCP/AWS**—and automated correlation, matchback, and feature pipelines, **cutting ops cost by 25%**.
- Price & Promo elasticity models **improved revenue efficiency by ∼15%**; translated model outputs into **budget scenarios** that shaped executive spend decisions.
- **Led R&D strategy** and authored technical documentation that **secured SR&ED tax-credit funding**, directly financing the company's applied research and platform roadmap.
- *Tech stack:* Python, SQL, Google Meridian, TensorFlow/TFP, FastAPI, Celery, Redis, BigQuery, AWS (S3, Lambda), GCP (Cloud Run, GCE, GCS), Docker, GitHub Actions, Firebase/Firestore, Looker Studio

Software Engineer (Data Infrastructure) | Spire Global Inc. | *ON, Canada* *Nov 2022 - Feb 2023*

- Built **production backends** on **satellite + RF maritime** data for vessel detection and **global shipping intelligence** at scale.
- Shipped scalable geospatial APIs (**GIS**, OGC Web Map) powering real-time spatial analytics.
- *Tech stack:* Python, Go, AWS, GraphQL, GIS, Image Processing

Backend Software Developer | Haqdarshak | *MH, India* *Jan 2021 - Oct 2021*

- Shipped microservices/APIs supporting **10,000+ daily users**, including high-volume transactional paths for India's COVID platform.
- **Cut retrieval latency by 40%** via PostgreSQL indexing, query plans, and safer migrations under traffic spikes.
- Owned backend for welfare/fintech-like claim flows—REST on **Django**—improving reliability under government-scheme traffic.
- *Tech stack:* Python, Django, REST APIs, PostgreSQL, Docker, Kubernetes, AWS

Software Engineer | RecruitNet | *Hong Kong SAR* *Jun 2020 - Dec 2020*

- Built ETL/scraping pipelines at scale; applied **NLP** to unstructured hiring data to improve candidate matching.
- Applied **NLP and text analytics** to unstructured hiring data to improve candidate matching and recruitment analytics.
- *Tech stack:* Python, Java, Selenium, BeautifulSoup, NLP

TECHNICAL SKILLS

ML & modeling: Bayesian modeling (Google Meridian), predictive modeling, regression, feature engineering, TensorFlow/TFP, NLP
Data & MLOps: SQL, BigQuery, PostgreSQL, Redis, Spark, Kafka; Celery, Docker, CI/CD; Looker Studio, Tableau
Software & cloud: Python, Go, Java, TypeScript; FastAPI, Django, REST/GraphQL; AWS (S3, EC2, Lambda), GCP (Cloud Run, GCS)

EDUCATION

MSc. in Big Data Analytics - *Trent University (GPA: 4.0/4.0)* *May 2021 - Sept 2022*

- Thesis research: compared deep learning sequence models (C-BiLSTM vs. Simple BiLSTM) for predictive tasks. *ON, Canada*

BE. in Computer Engineering - *Gujarat Technological University (GPA: 3.9/4.0)* *Aug 2016 - Sept 2020*

- Design Lead at Computer Society of India SB, Mentor of Coding & Hackathon Club *GJ, India*

PROJECTS

Retail Optimization for Behavior Analysis & In-Store Navigation *Python, Django, PostgreSQL, React, NLP, TensorFlow*

- Built **predictive analytics** for forecasting, assortment, and personalization from large-scale customer/sales data; modeled in-store behavior for merchandising experiments.

CERTIFICATIONS & AWARDS

- **Google Meridian Partner Certification** — certified in Google's open-source Bayesian MMM framework.
- **Employee of the Year (2023)**, Miix Analytics — high-impact platform and client outcomes.
- **2024 Global 30-Under-30**, Significant Insights — excellence in research, data, and insights.
- Published deep learning / NLP articles (NMT, Image Super-Resolution) with **30,000+ combined views**.